

PEREIRA, Colombia

WMO: 802100

Lat: 4.813N

Lon: 75.740W

Elev: 1346

StdP: 86.17

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	16.8	17.1	13.2	11.2	22.1	14.2	11.9	21.0	5.8	21.6	5.2	21.3	1.8	90	0.227

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	30.1	21.1	29.2	21.2	28.4	21.2	22.7	27.3	22.2	27.0	21.9	26.8	2.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.7	24.8	21.0	18.5	24.7	20.3	17.7	24.2	74.6	27.7	73.1	27.3	71.4	26.8	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 5.2	4.5	4.1	DB	15.5	32.0	0.7	1.5	14.9	33.1	14.5	33.9	14.1	34.8	13.6	35.9	(4)
(5)			WB	13.7	23.7	0.8	1.0	13.1	24.4	12.6	25.0	12.2	25.5	11.6	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	22.8	23.1	22.7	22.4	22.3	22.5	22.6	22.8	22.8	22.0	21.9	22.5	(6)	
	DBStd	1.22	1.28	1.23	1.36	1.25	1.13	1.12	1.13	1.19	1.18	1.08	1.02	1.13	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	4578	397	366	394	373	382	375	392	398	383	373	358	387	(10)	
	CDD18.3	1537	139	132	136	123	124	125	134	139	133	115	108	129	(11)	
	CDH23.3	5997	643	601	544	426	396	461	564	639	598	358	298	469	(12)	
	CDH26.7	881	124	118	87	55	39	54	78	114	113	27	16	54	(13)	
(14) Wind	WSAvg	2.0	2.0	2.0	2.0	1.9	1.9	1.9	1.9	2.0	2.0	2.0	1.9	2.0	(14)	
(15) Precipitation	PrecAvg	2398	139	164	228	270	245	160	130	126	162	296	286	190	(15)	
	PrecMax	3199	282	308	341	413	411	261	243	265	301	433	463	341	(16)	
	PrecMin	1664	50	72	101	149	139	29	32	25	86	156	189	30	(17)	
	PrecStd	387	62	67	66	64	62	61	54	52	46	66	75	74	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.6	30.9	30.2	29.7	29.1	29.9	30.0	30.8	30.8	28.5	28.0	29.1	(19)	
		MCWB	21.1	21.2	21.8	22.1	22.1	20.8	20.9	20.3	20.8	21.4	21.6	21.4	(20)	
	2%	DB	29.1	29.2	28.9	28.1	27.9	28.0	28.4	29.1	29.1	27.2	27.0	28.0	(21)	
		MCWB	21.0	20.8	21.5	21.8	21.9	21.4	21.0	20.9	21.0	20.9	21.1	21.3	(22)	
	5%	DB	28.1	28.1	27.8	27.0	26.8	27.0	27.2	27.9	28.0	26.2	26.1	27.1	(23)	
		MCWB	20.8	20.9	21.3	21.3	21.4	21.2	21.0	20.9	21.0	20.7	20.8	21.0	(24)	
	10%	DB	27.0	27.0	26.3	25.9	25.8	26.0	26.2	26.8	26.8	25.4	25.1	26.0	(25)	
		MCWB	20.5	20.7	20.8	20.9	21.0	21.0	20.7	20.7	20.7	20.4	20.4	20.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	22.7	22.9	23.0	23.1	23.1	22.5	22.3	22.3	22.4	22.4	22.5	(27)	
		MCDB	28.2	28.4	27.7	27.4	27.4	26.8	27.3	27.8	27.9	26.9	26.3	27.0	(28)	
	2%	WB	21.6	22.0	22.1	22.3	22.3	22.2	21.8	21.7	21.7	21.6	21.6	21.8	(29)	
		MCDB	27.4	27.6	27.1	26.7	26.5	26.4	26.7	27.2	27.3	26.1	25.8	26.4	(30)	
	5%	WB	21.2	21.5	21.6	21.7	21.7	21.6	21.2	21.2	21.2	21.1	21.1	21.4	(31)	
		MCDB	26.7	26.6	26.3	26.0	25.7	25.8	26.0	26.5	26.7	25.3	25.2	25.9	(32)	
	10%	WB	20.7	21.0	21.1	21.1	21.1	21.1	20.9	20.8	20.8	20.6	20.6	20.9	(33)	
		MCDB	25.8	25.7	25.5	25.1	24.9	25.1	25.5	25.8	25.9	24.6	24.4	25.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	8.4	8.0	7.6	8.0	8.6	9.0	9.1	8.1	7.8	8.2	(35)	
		MCDBR	10.6	10.2	9.9	9.4	8.9	9.2	9.7	10.4	10.7	9.0	8.7	9.3	(36)	
	5% WB	MCWBR	4.6	4.3	4.2	4.5	4.4	4.4	4.5	4.8	4.8	4.3	4.1	4.4	(37)	
		MCWBR	9.3	9.1	8.9	8.6	8.2	8.6	9.1	9.5	9.7	8.6	8.3	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.425	0.458	0.485	0.449	0.404	0.381	0.384	0.412	0.424	0.436	0.417	0.415	(40)	
		taud	2.317	2.216	2.140	2.263	2.406	2.475	2.461	2.358	2.316	2.299	2.364	2.365	(41)	
	Ebn at Noon	Ebn at Noon	895	872	847	862	882	892	893	883	887	881	896	898	(42)	
		Edh at Noon	132	149	162	140	118	108	111	126	134	136	125	124	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.35	4.34	4.18	4.20	4.33	4.45	4.64	4.67	4.67	4.33	4.12	4.14	(44)	
	RadStd	RadStd	0.34	0.39	0.27	0.21	0.19	0.17	0.21	0.23	0.25	0.17	0.18	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	+0.34	+0.32	N/A	N/A	N/A	N/A	N/A	-185	+184	N/A

Nomenclature: See separate page